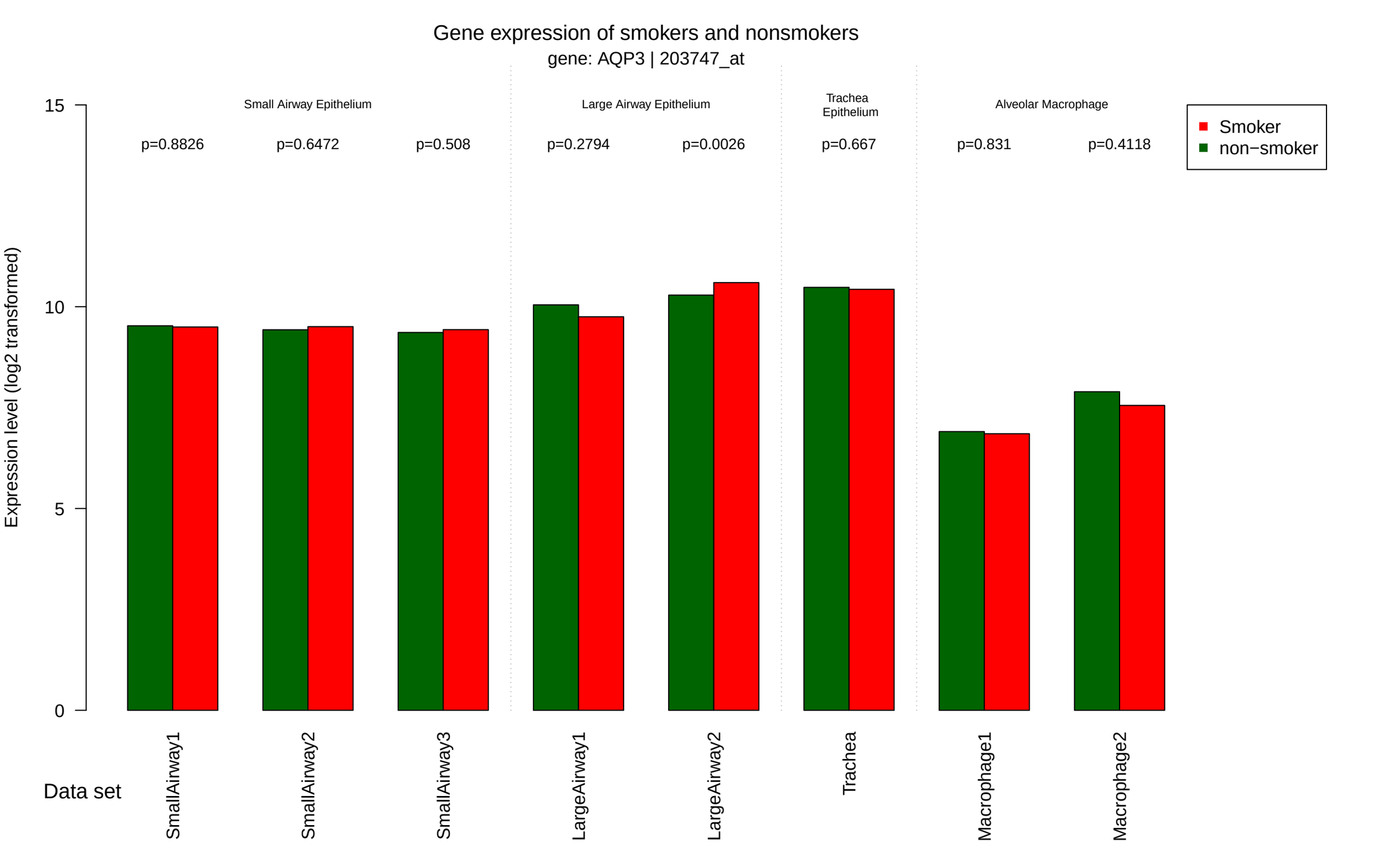
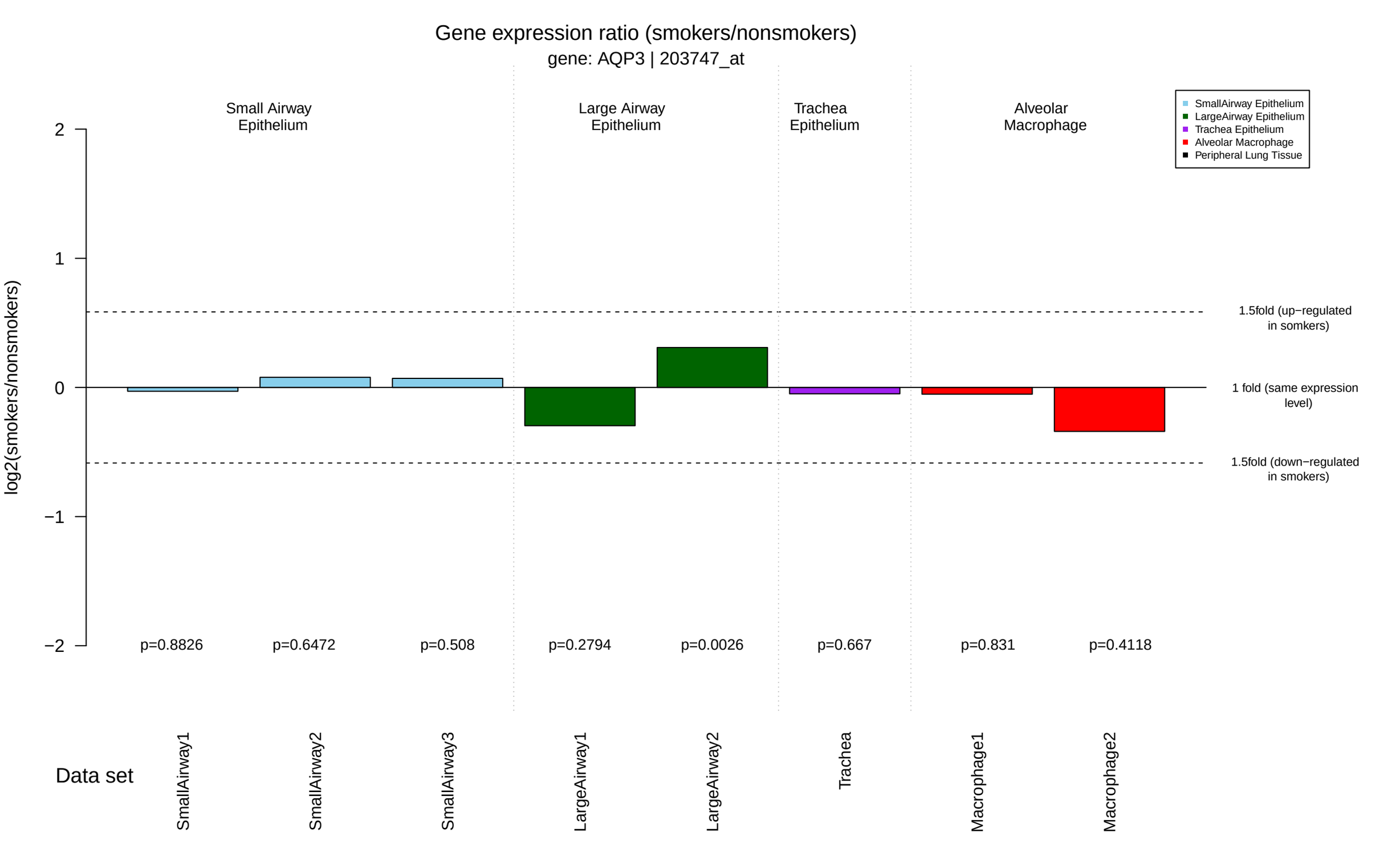
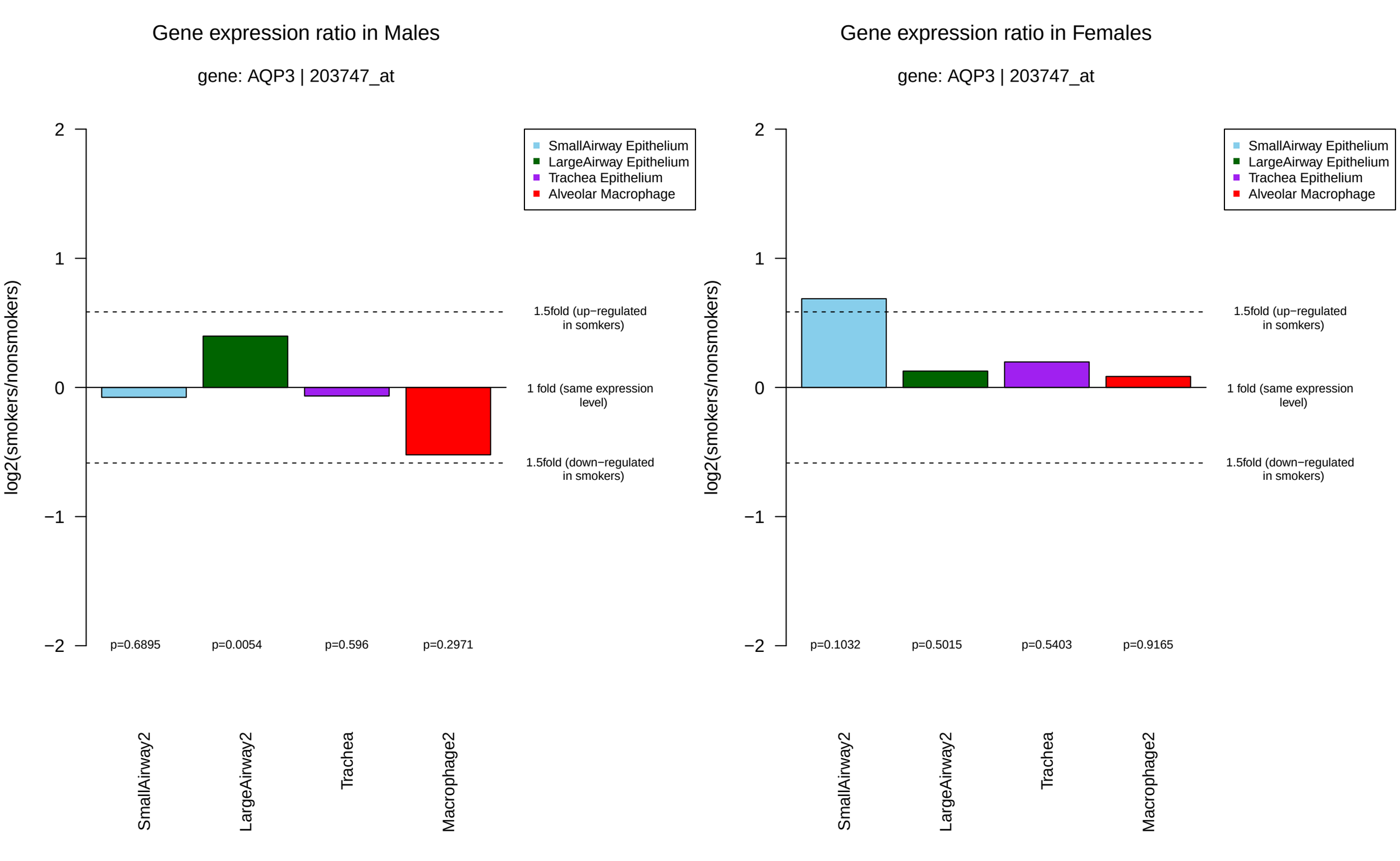
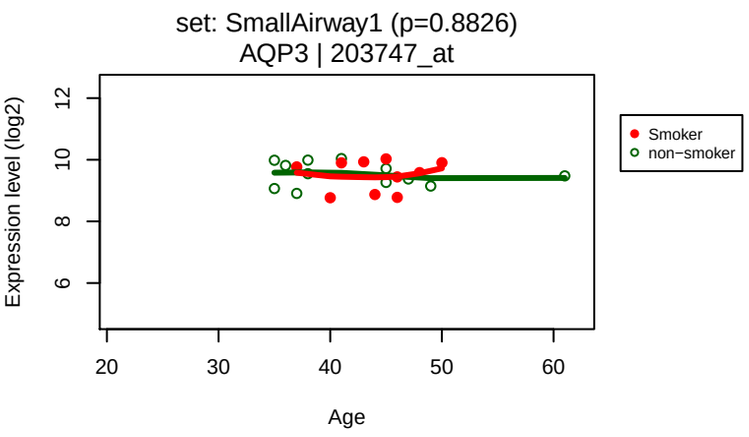
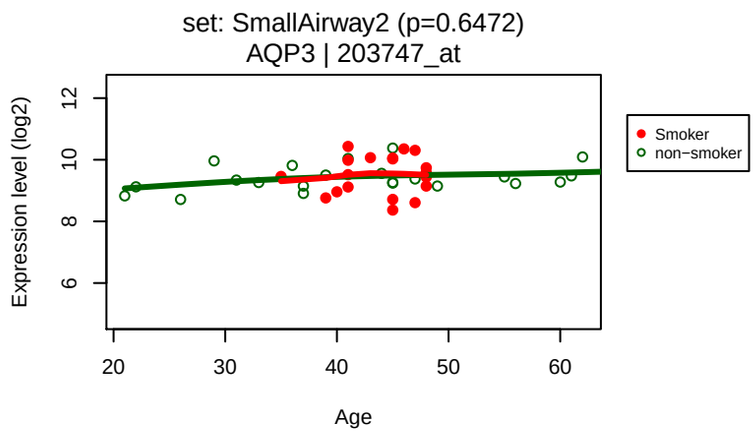
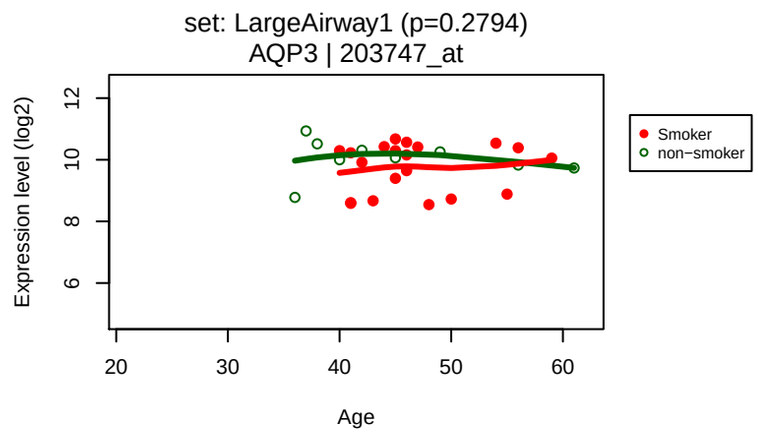

Gene expression of smokers and nonsmokers
gene: AQP3 | 203747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8826
p=0.6472
p=0.508
p=0.2794
p=0.0026
p=0.667
p=0.831
p=0.4118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

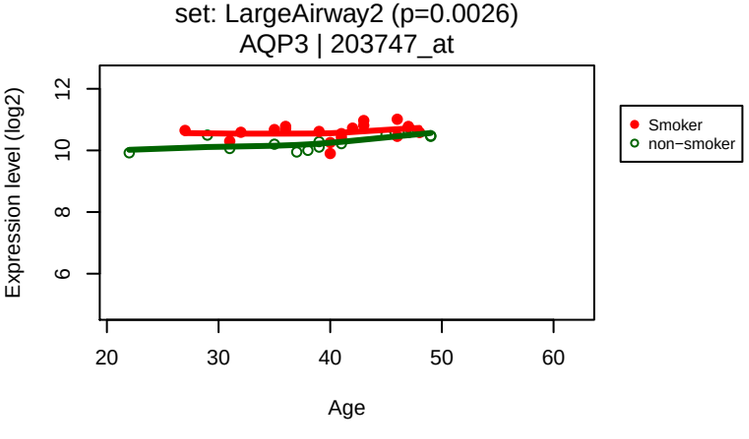

Gene expression ratio (smokers/nonsmokers)
gene: AQP3 | 203747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8826
p=0.6472
p=0.508
p=0.2794
p=0.0026
p=0.667
p=0.831
p=0.4118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

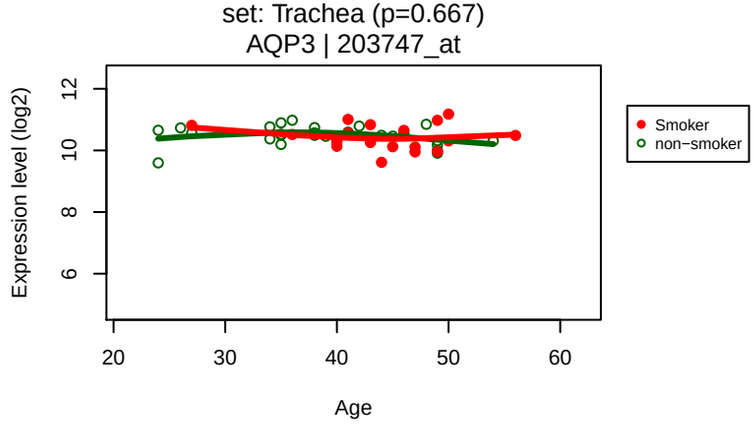

Gene expression ratio in Males
gene: AQP3 | 203747_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6895
p=0.0054
p=0.596
p=0.2971
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP3 | 203747_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1032
p=0.5015
p=0.5403
p=0.9165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

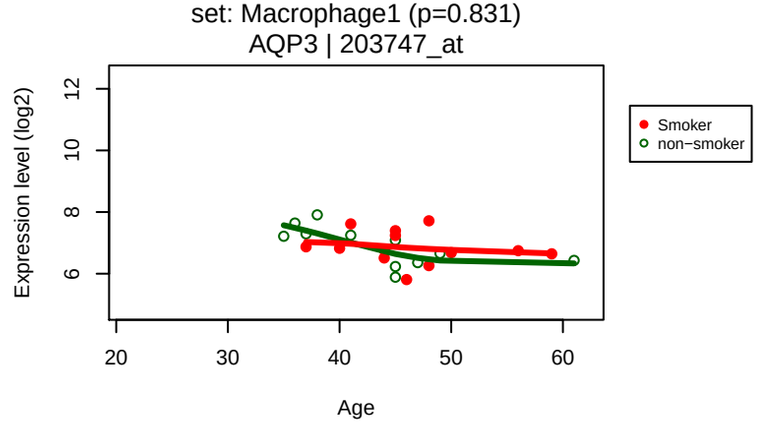

set: SmallAirway1 (p=0.8826)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker

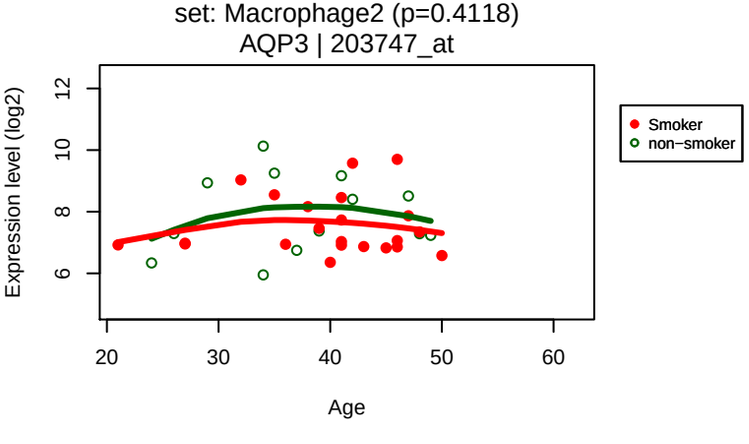

set: SmallAirway2 (p=0.6472)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker

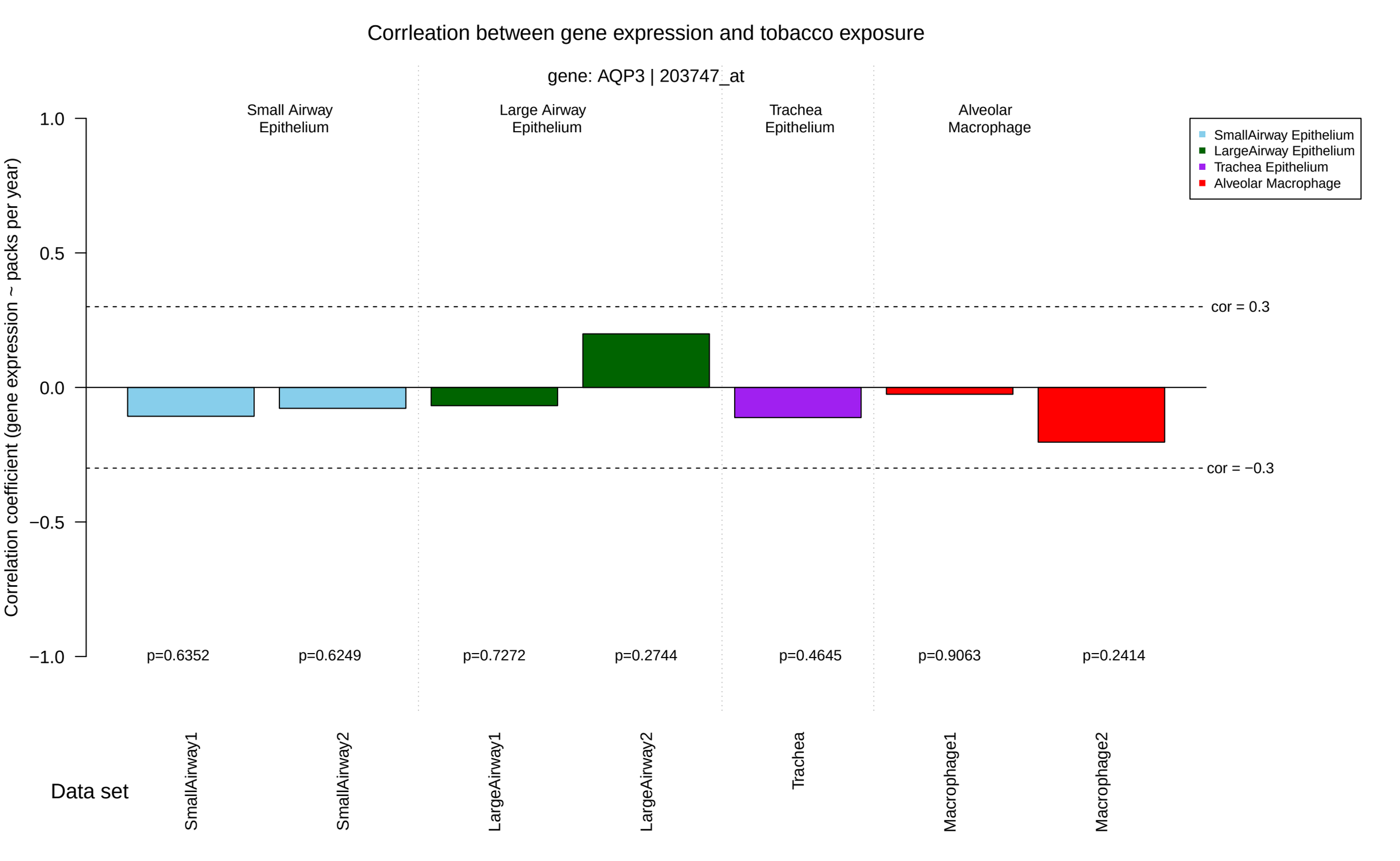

set: LargeAirway1 (p=0.2794)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0026)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.667)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.831)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4118)
AQP3 | 203747_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP3 | 203747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6352
p=0.6249
p=0.7272
p=0.2744
p=0.4645
p=0.9063
p=0.2414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2